

MATH3171 (Mathematical Biology III)
YEAR 2025–26, EASTER TERM

RESIT VIVA

1. (a) Consider the following models for the population of a single species, $x(t)$, evolving over time, t . Each model depends on a parameter, $k > 0$. For each model, provide a short biological interpretation and state any permissible equilibria and their stabilities. It may help to draw a phase portrait of dx/dt against x in each case.
 - (i) $\frac{dx}{dt} = kx$,
 - (ii) $\frac{dx}{dt} = x(1 - x) - k$,
 - (iii) $\frac{dx}{dt} = -x(x + k)(x - k)$.
- (b) Draw a bifurcation diagram of permissible equilibria, x_0 , against k for model (ii) above, indicating stability appropriately on the diagram.

2. (a) In words, describe what is meant by a *Turing instability*. Why are these instabilities potentially important in biology?
- (b) For each of the following models of populations, give a brief explanation why they can or can not have Turing instabilities. Clearly state any conditions on functions or parameters necessary to generate a Turing instability for the spatial domain $x \in [0, L]$, where $L \gg 1$ is arbitrarily large (i.e. you can approximate the domain size L as infinite).
 - (i) $\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + f(u)$,
 - (ii) $\frac{\partial u}{\partial t} = -a \frac{\partial^2 u}{\partial x^2} - \frac{\partial^4 u}{\partial x^4} + f(u)$.